966445

Course Name: **Design and Analysis of Algorithms**

Course Outcome

CO1: Understanding of complexity representation in terms of Big Oh, Theta and Omega notations.
CO2: Derive and solve recurrences describing the performance of divide-and-conquer algorithms (quick sort and merge sort).
CO3: Compare and analyze different data structures (RB Tree, B Tree, Binomial Heaps, Fibonacci Heaps).
CO4: Understand the major graph algorithms (DFS, BFS, Dijkstra's Bellman Ford) and their analyses.
CO5: Understand the greedy paradigm and able to analyze when an algorithmic design situation calls for it. Synthesize greedy algorithms (Optimal Reliability Allocation, Minimum Spanning Trees, factorial Knapsack) and analyze them.
CO6: Synthesize dynamic-programming algorithms (0/1 knapsack problem, Resource allocation problem, Warshal's and Floyd's algorithms) and analyze them.
CO7: Understand the backtracking paradigm and able to analysis when an algorithmic design situation calls for it. Synthesize backtracking algorithms (N Queen Problem, TSP Problem, sum of subsets problem, Graph Coloring) and analyze them.
CO8: Understand the branch and bound paradigm and able to analysis when an algorithmic design situation calls for it. Synthesize branch and bound algorithms (N Queen Problem, TSP Problem, Hamiltonian Cycles, Graph Coloring) and analyze them.

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Mid Term Examination, Even Semester 2024-25
B.Tech(CS / IIoT), 3rd Year, Semester-VI
BCSC 1012 & Design and Analysis of Algorithms

Time: 2 Hours **Maximum Marks: 30**

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Recurrence relation of Quick sort is T(n)=T(k-1)+T(n-k)+partition function time; where n is the number of elements in array and k is the position where the pivot element is stored. Let partition function of quick sort takes $O(\sqrt{n})$ time then what are the recurrence relations of Quick sort in best-case and worst-case. Solve recurrence relations of best-case and worst-case according to mentioned partition function time.	3	2	A	C
2	Let you have given Red-Black tree and B-Tree, then which one is better for insertion and searching operations. Is Red-Black tree a variation of B-Tree, if yes then justify your answer.	3	3	An	C
3	Apply the procedure of Insertion sort and Shell sort on a given sequence of elements to sort in increasing order , where elements are 5, 7, 2, 10, 12, 9, 15, and 1.	3	2	An	P

	Find the number of swaps required Insertion sort and Shell sort according to given elements.				

4	Given a part of a program.
	int s=0;
	for(int i= 1 ; i<=n; i=i*3){
	for(int j= 1 ; j<=i; j++){
	s=s + n ; }
	}
	Is it true that value of variable s is $\Theta(n^2)$ after execution of code, if yes then how; if not then what will be the value of s in theta notation.
		3	1	An	C

5	Let an array a[n] contains floating point values in between 10^{-3} to 10^{3}. A number contains at most 3 decimal point values like 2.345 or 0.12 or 23.567 but not 3.2356 or 56.78543(last two numbers are not part of array because it contains more than 3 decimal point values). How Bucket sort and Radix sort can be modified so that it can be sorted using Bucket sort and Radix sort .	3	2	C	M

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Insert 10 elements in a sequence 2,3,4,5,6,7,1,10,11,13 in an empty B-tree of degree 3. After insertion, delete elements in a sequence 2 and 5 from B tree. Design tree after each insertion and deletion operation.	5	3	An	F
7	Write an algorithm of Heap sort, and find time complexity in worst-case. Apply Heap sort procedure in a given sequence of elements: 21, 45,72, 33, and 26.	5	2	A	F
8	Let there are two Heaps X and Y having m and n elements respectively then write an algorithm such that it can be merged into one Heap in two cases Case1: If X and Y are Binary Min Heap. Case2: If X and Y are Binomial Min Heap. Find worst case time complexity for these two cases.	5	3	E	P